LIKHITHA BANDARU

Software Engineer | Data & AI | Python | SQL | LLMs

Bangalore / Tirupati, India | +91 8341950658 | likhithabandaru2005@gmail.com | LinkedIn | GitHub

SUMMARY

Software Engineer (B.Tech IT, 2026) and AWS Certified Cloud Practitioner with solid software, data, and AI/ML engineering skills. Builds data products and intelligent applications using Python, advanced SQL, ETL/ELT data pipelines, and large language models (LLMs) such as Gemini and Claude. Experienced with Snowflake, dbt, Airflow, CI/CD, and MLOps, with a focus on data quality, metadata, data governance, and compliance. Collaborative, proactive, and comfortable in Agile and open source / InnerSource workflows.

TECHNICAL SKILLS

Programming: Python (preferred), SQL (advanced), Java, Go (basic), JavaScript
Data Engineering: ETL/ELT pipelines, data products, data modeling, Snowflake, dbt, Apache Airflow / Astronomer, Fivetran, data warehouse, custom data adapters
AI / ML: Large Language Models (LLMs), generative AI, RAG, prompt engineering, Gemini, Claude, Codex, open-source LLMs, LangChain, embeddings, Scikit-learn, Pandas, NumPy
MLOps & DevOps: MLflow, model registry, CI/CD (GitHub Actions), Docker, Git, pytest, AWS
Data Governance: Metadata, data catalog, data classification and tagging, data masking, role-based access control, data quality, SLOs, compliance
Architecture & Practices: Data mesh principles, InnerSource / open source, Agile, REST APIs (Flask), software engineering best practices

PROJECTS

SmartData Assist - LLM-Powered Data Product Assistant GitHub | Live Demo
Tech stack: *Python, Gemini API, Claude API, LangChain, FAISS, Flask, PostgreSQL, Docker*

- Developed and deployed a generative AI application using large language models (LLMs) - Gemini and Claude - that answers business questions over data products with retrieval-augmented generation (RAG).
- Built metadata-driven retrieval: data product descriptions, schemas, and ownership tags are embedded in a vector index so the LLM returns grounded, cited answers.
- Generated SQL from natural language with guardrails (read-only queries, schema validation) and masked PII columns before results reach the model.
- Containerized with Docker and shipped via a GitHub Actions CI/CD pipeline with pytest tests and prompt evaluation checks.

DataPulse - ELT Data Pipeline and Data Products GitHub | Live Demo
Tech stack: *Python, SQL, Airflow, dbt, Snowflake, Fivetran-style connectors, Great Expectations*

- Migrated a script-based ingestion flow to an ETL/ELT model orchestrated with Apache Airflow, loading source data into a Snowflake warehouse.
- Separated source-aligned data products from aggregate data products using dbt layers (staging, intermediate, marts) and advanced SQL transformations (CTEs, window functions).
- Wrote custom Python adapters to ingest REST API and CSV sources; standardized common identifiers and keys so data products join cleanly.
- Added data quality tests, freshness SLOs, data catalog documentation, classification tags, and role-based access with column masking for governance and compliance.

InsightML - Predictive Analytics with MLOps GitHub | Live Demo
Tech stack: *Python, Pandas, Scikit-learn, MLflow, Flask, Docker, GitHub Actions*

- Built a feature pipeline and trained classification models, comparing them with precision, recall, and F1-score.
- Tracked experiments, metrics, and model versions in MLflow and registered the best model in a model registry for reuse.
- Served predictions through a Flask REST API with input validation; automated testing and deployment with CI/CD following MLOps practices.
- Built a dashboard explaining results and feature importance to non-technical business users.

EDUCATION

B.Tech, Information Technology - Sri Venkateswara College of Engineering (SVCE), Tirupati | CGPA: 8.7/10 2022 - 2026
Intermediate (MPC) - Narayana Junior College | CGPA: 8.5/10 2020 - 2022
Coursework: Machine Learning, DBMS, Data Structures & Algorithms, Software Engineering, OOP, Operating Systems

CERTIFICATIONS

- AWS Certified Cloud Practitioner - Amazon Web Services (Oct 2025)
- Tech Mahindra Nandi Certification - Tech Mahindra

ADDITIONAL INFORMATION

Strengths: Collaboration, quality focus, user empathy, prioritization, bias for action | Languages: English, Telugu, Hindi